**Control Estadístico de la Calidad**

Examen Argumentativo

Nombre: ________________________________________ Matrícula:___________________________

**Instrucciones:** Muestra cómo obtienes tu respuesta. Puedes usar Excel o Minitab para tus cálculos de probabilidades o percentiles, pero no uses otros archivos o hagas uso de la IA. Respeta el código de ética del curso y del Tecnológico de Monterrey.

1. A Shewhart $\bar{X}$ chart is used with subgroup size $n = 6$. The process has $\bar{\bar{X}} = 55$ and $\bar{R} = 6.5$.

   Calculate the upper control limit (UCL) of the chart.

2. A process has specification limits LSL $= 81$ and USL $= 105$. The estimated process standard deviation is $\sigma = 4.5$.

   Calculate the potential capability index $C_p$.

3. A Shewhart control chart has center line 118, UCL $= 128$ and LCL $= 103$. A new sample statistic is 129.

   What is the best conclusion?

   (a) The process variability has decreased.
   (b) No action is needed; the point is within limits.
   (c) The process is centered and capable.
   (d) The process is out of statistical control.

4. In an inspection of 200 units, 14 defective units were found. The fraction defective is $p = 0.07$.

   Which interpretation is correct?

   (a) The process capability index equals 0.07.
   (b) Exactly 93% of future units will be conforming.
   (c) Approximately 7% of the inspected units were defective.
   (d) The process is automatically in statistical control.

5. A process has $C_p = 1.4$ and $C_{pk} = 0.9$.

   Which conclusion is most appropriate?

   (a) The process spread is adequate, but the process is not perfectly centered.
   (b) The specification limits are wider than the process spread because $C_{pk} > C_p$.
   (c) The process is out of statistical control.
   (d) The process is perfectly centered between specifications.